

Algebra II with Trigonometry

Chapter 5.2

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Basic Tangent Ratio In a right triangle, if the side opposite to acute angle θ is 3 units long and the adjacent side is 4 units long, what is the value of $\tan(\theta)$?

Reciprocal Functions If you know that $\cos(\theta) = \frac{1}{2}$, what is the value of its reciprocal function, $\sec(\theta)$?

Special Right Triangles (45-45-90) In a 45° - 45° - 90° special right triangle, if the lengths of the two legs are both 1, what is the exact length of the hypotenuse?

Radians to Degrees The special values table lists $\frac{\pi}{3}$ radians. What is the equivalent measure of this angle in degrees?

Evaluating Special Angles (Sine) Based on the special right triangles, what is the exact numerical value of $\sin(30^\circ)$?

Evaluating Special Angles (Tangent) Without using a calculator, what is the exact value of $\tan(45^\circ)$?

Line of Sight Terminology If an observer looks straight ahead horizontally and then tilts their eyes downward to look at an object below, what is the specific term for the angle formed between the horizontal and their line of sight?

Mental Pythagorean Theorem & Sine In a right triangle, if the side opposite to angle θ is 3 and the adjacent side is 4, you can deduce the hypotenuse is 5. What is the value of $\sin(\theta)$?

Evaluating Simple Trig Expressions Using your knowledge of special angles, what is the sum of $\sin(30^\circ)$ and $\cos(60^\circ)$?

Reciprocal Relationships If a math problem states that $\tan(\theta) = \frac{5}{3}$, what is the exact value of $\cot(\theta)$?